Individual Participant Data Meta-Analysis

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Introduction

Individual participant data (IPD) meta-analysis pools raw participant-level data from multiple studies, enabling analyses that aggregate-data meta-analyses cannot – time-to-event across studies, individual-level effect modification, and harmonised outcome definitions. It is the gold standard when feasible but requires data-sharing agreements.

Prerequisites

Meta-analysis pooling; mixed-effects models.

Theory

Two-stage: analyse each study separately to obtain effect and variance; pool with standard meta-analysis methods.

One-stage: fit a single hierarchical model on the combined data with study as a random effect. Allows individual-participant-level interactions.

Both approaches typically yield similar results for main effects; one-stage is preferred for individual-level moderator analyses.

Assumptions

Data-sharing agreements in place; common outcome definitions across studies; correct handling of study-level heterogeneity.

R Implementation

```
library(lme4)

set.seed(2026)
n_studies <- 5
participants_per <- 100

# Simulate IPD
df <- do.call(rbind, lapply(1:n_studies, function(s) {
  n <- participants_per
  data.frame(
    study = s,
    arm = factor(sample(c("trt", "ctrl"), n, replace = TRUE)),
    age = rnorm(n, 50, 10)
  )
}))
df$arm_num <- as.integer(df$arm == "trt")
df$y <- 0.4 * df$arm_num + 0.05 * df$age +
```

```r
        rnorm(nrow(df), 0, 1) +
        rep(rnorm(n_studies, 0, 0.3), each = participants_per)

# One-stage mixed-effects model
one_stage <- lmer(y ~ arm + age + (1 | study), data = df)
summary(one_stage)$coefficients

# Two-stage: per-study effects, then pool
per_study <- split(df, df$study) |> sapply(function(s) {
  m <- lm(y ~ arm + age, data = s)
  c(est = coef(m)["armtrt"], var = vcov(m)["armtrt", "armtrt"])
})
library(metafor)
ma <- rma(yi = per_study["est.armtrt", ], vi = per_study["var", ],
          method = "REML")
ma$b
```

Output & Results

One-stage arm coefficient and the equivalent two-stage pooled estimate; typically close agreement for simple models.

Interpretation

"IPD meta-analysis estimated the intervention effect at 0.42 (95 % CI 0.28-0.56, $p < 0.001$); two-stage and one-stage approaches agreed closely, supporting the finding's robustness to analytic framing."

Practical Tips

- Prospective IPD consortium studies avoid selection bias from data-availability issues.
- Harmonise outcomes before pooling; different outcome definitions across studies cause bias.
- One-stage model allows individual-level interactions (effect modifiers); more valuable than ecological moderators.
- For survival endpoints, IPD is required (aggregate hazard ratios are difficult to pool correctly).
- Report PRISMA-IPD checklist for transparency.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis